

WEB PROGRAMMING SECURITY

JAVASCRIPT-PART3 & DOM

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WHAT IS THE DOM?

Definition

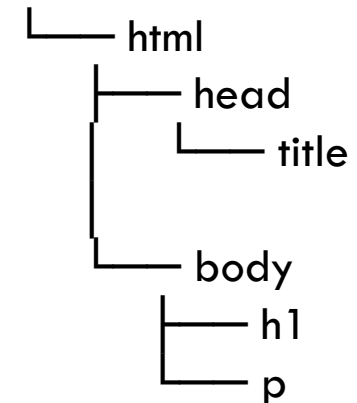
DOM - Document Object Model

It is a programming interface that represents the HTML page as objects that JavaScript can access and modify.

When a browser opens a webpage it does three main steps.

- Step 1: Browser reads HTML file
- Step 2 : Browser converts HTML into DOM Tree
- Step 3 : JavaScript can access the DOM

DOM Tree becomes
document



WHY THE DOM IS IMPORTANT

- Without DOM JavaScript cannot interact with the page.
- With DOM JavaScript can :
 - change text
 - change color
 - read input
 - create elements
 - delete elements

EXAMPLE

HTML

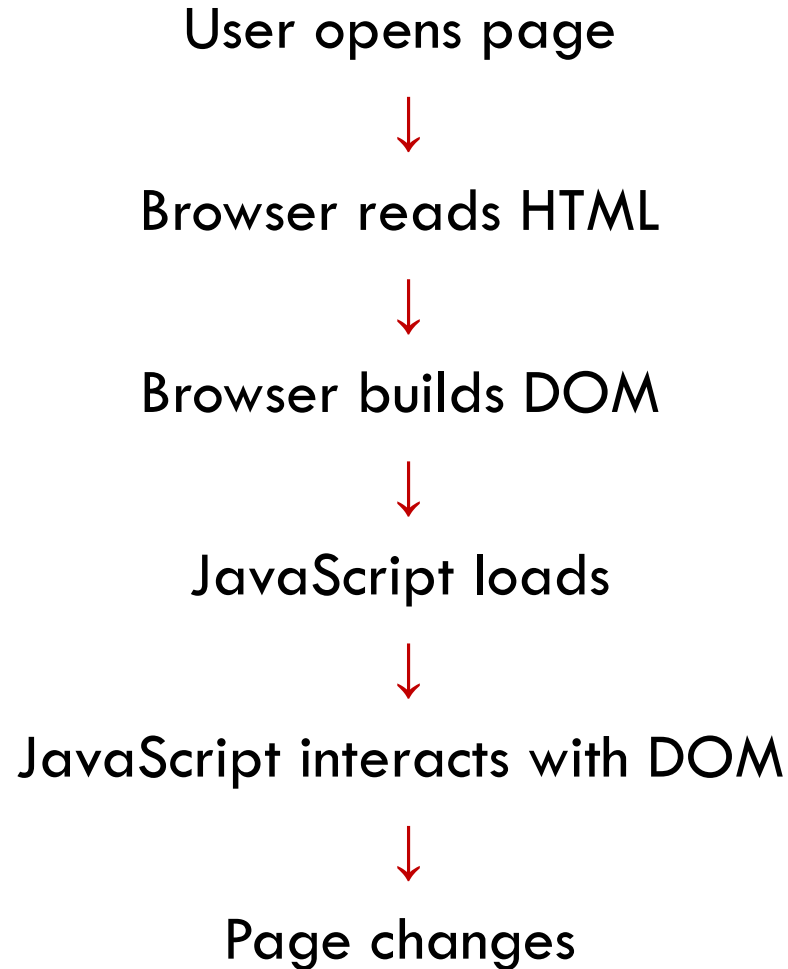
```
<h1 id="title">Original Text</h1>
```

JavaScript

```
document.getElementById("title").innerHTML = "New Text"
```

HOW JAVASCRIPT EXECUTES IN BROWSER

The execution flow is:



PROCEDURE

Step 1 : Create Project Folder

Step 2 : Write HTML

```
<!DOCTYPE html>

<html>

<head>

<title>DOM Example</title>

</head>

<body>

<h1 id="title">Hello Students</h1>

<button onclick="changeText()">Change Text</button>

<script src="script.js"></script>

</body>

</html>
```

Step 3: Write JavaScript

Open script.js

```
function changeText() {

    document.getElementById("title").innerHTML =
    "Text changed by JavaScript"

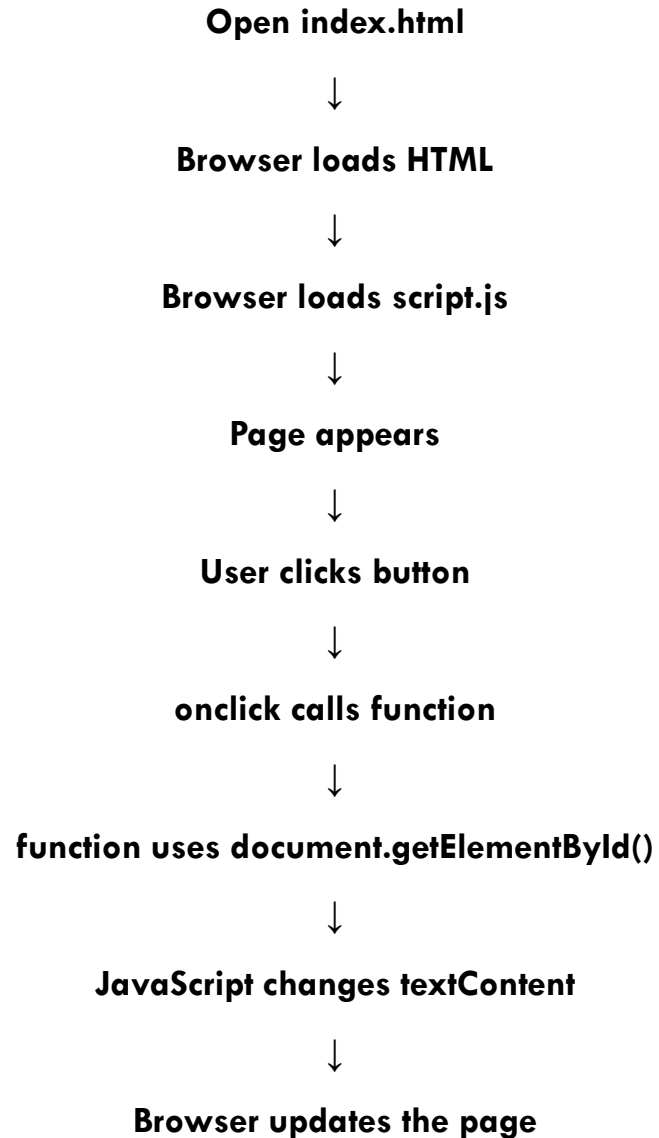
}
```

Step 4 : Run the Code

- Save files
- Double click index.html

FLOW CHART FOR TEXT CHANGING

Button changes text



EXAMPLE (SIMPLE CALCULATOR)

HTML

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Calculator</title>
</head>
<body>
  <h1>Simple Calculator</h1>
  <input type="number" id="num1" placeholder="First number">
  <input type="number" id="num2" placeholder="Second number">
  <button onclick="addNumbers()">Add</button>
  <p id="result">Result appears here</p>
  <script src="script.js"></script>
</body>
</html>
```

JavaScript

```
function addNumbers() {
  let first = document.getElementById("num1").value
  let second = document.getElementById("num2").value

  let sum = Number(first) + Number(second)

  document.getElementById("result").textContent = "Result = " + sum
}
```


GREETING APPLICATION

HTML

```
<input type="text" id="studentName">  
<button onclick="welcome()">Welcome</button>  
<p id="message"></p>
```

JavaScript

```
function welcome() {  
    let name = document.getElementById("studentName").value  
    document.getElementById("message").innerHTML =  
        "Welcome " + name  
}
```

The DOM lets javascript access element properties and styles, so changing style through javascript is part of DOM manipulation

HTML

```
<!DOCTYPE html>
<html>
<head>
  <title>Change Style</title>
</head>
<body>
  <h1 id="title">DOM Styling Example</h1>
  <button onclick="changeStyle()">Change Style</button>
  <script src="script.js"></script>
</body>
</html>
```

JavaScript

```
function changeStyle() {
  let heading = document.getElementById("title")
  heading.textContent = "Style Changed"
  heading.style.color = "red"
  heading.style.backgroundColor = "yellow"
}
```

HOME WORK

- modify the calculator to do → **Subtraction** then **Multiply**